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Deep Learning-Driven Mechanism Prediction and New Electrocatalyst Generation for Metal-Air Batteries

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Abstract

In this work, copper-doped Electrocatalyst was synthesized via pyrolysis, and the electrocatalyst was characterized by XRD, FTIR, and XPS, followed by electrochemical testing. Also, a new electrocatalyst for metal-air batteries was designed, and reaction processes are predicted using large language models (LLMs) such as ChemBERT and ChemGPT. To train the models, RDKit was used to process SMILES representations of functional groups, doping techniques, and electrolyte media. While ChemGPT produced new electrocatalysts with an accuracy of 0.83, ChemBERT, which had been optimized to 0.8, was used to predict mechanisms. To speed up catalyst identification, the methodology incorporates reaction pathway predictions.

Keywords: Electrocatalyst, Prediction, Energy, Sustainability

Introduction

High global demand for clean, renewable, and sustainable energy requires proton exchange membrane fuel cell technology and metal-air battery (She *et al.* 2017; Sun *et al.* 2014). In the pursuit of more sustainable energy solutions, metal-air batteries, particularly Zinc-air batteries, offer a promising alternative to conventional energy conversion systems. However, their widespread adoption has been limited by the sluggish kinetics of redox reactions at the cathode, despite advantages such as high energy density, intrinsic safety, and cost-effectiveness (Kundu *et al.* 2021; Li *et al.* 2020).

Platinum is widely used in metal-air batteries. However, it is a very expensive metal, thereby increasing the cost of production for industries. There is a high need to substitute platinum with non-precious metals. Many attempts were made to replace platinum-based electrocatalysts with nonprecious and cheaper metals for the oxygen reduction reaction ORR which is a very crucial process in metal-air batteries (Nørskov *et al.* 2004; Liu *et al.* 2020). Metallic doping using metals such as Fe, Co, or Ni, etc, can be achieved by atomically dispersing the element into the carbon framework (Khattak *et al.* 2022; Rojas-Carbonell *et al.* 2017).

Metals can be doped into carbon via pyrolysis, which involves subjecting the precursors to high temperature (Serov *et al.* 2015). Metallic doped catalysts are effectively excellent and active in acidic conditions, interestingly exceptional in neutral conditions (Choi *et al.* 2015), and even better than Pt performance in alkaline conditions (Santoro *et al.* 2015).

The best and most effective material for ORR is transition metal and heteroatom Co-doped carbon-based materials (Jiao *et al.* 2013; Bhatt *et al.* 2020; Zhao *at al.* 2016). Doping transition metals in heteroatom-doped carbon can effectively boost the electrocatalytic activity of the material. Metals such as iron, copper, cobalt, and nickel have excellent electrocatalytic activity (Qiu *et al.* 2019). The metal-heteroatom moiety has been identified by many researchers to be the active site of the electrocatalyst (Gong *et al.* 2009; Qu *et al.* 2010; Lu *et al.* 2013).

Theoretical calculations predicted copper to have better ORR in comparison to other transition metals such as iron, cobalt, and nickel (Choi *et al.* 2012). The higher ORR of Copper could be due to its good electrical conductivity, which can enhance fast and effective electron transfer between reactants and active sites (Wu *et al.* 2020).

Due to the promising electrochemical properties of copper, it is vital to further investigate its activity and mechanism. Many researchers reported copper and Nitrogen co-doped carbon, which shows relatively good activity (Parse *et al.* 2021; Mathumba *et al.* 2020; Qin *et al.* 2019).

In this study, copper was co-doped with a phosphorus heteroatom to enhance the VR-700 (Vacume Residue) functionality and efficiency. Both the electrochemical and physical properties of the resulting composite were thoroughly investigated to evaluate its potential in electrocatalytic applications. Additionally, advanced deep learning techniques (ChemBERT and ChemGPT) were used to predict the underlying reaction mechanisms, providing further insights into the material's behavior and efficiency in electron transfer. This combined experimental and deep learning approach offers a comprehensive understanding of the system's performance and reactivity.

Material

Materials. Vacuum residue (VR) was collected from the local Ras Tanura refinery in the Eastern Province of Saudi Arabia. Phosphoric acid (85%) was purchased from Millipore-Sigma and used without further purification; Distilled water was processed from the Milli-Q (Milford, MA, USA) system.

Synthesis

Synthesis of the material was successfully conducted via pyrolysis. Vacuum residue (VR) was activated with phosphoric acid in an equal mass ratio. 10% copper (II) acetate was added to the mixture. The mixture was first heated at 100 °C for 5 minutes to enable the mixture to effectively combine. Then the mixture was activated from room temperature to a higher temperature of 700 °C gradually at the rate of 4.4 °C min^{-1}, and activated for another 2 hours. The second sample was synthesized in the same way but without copper at a temperature of 700 °C. The resulting Electrocatalysts were allowed to cool under an inert atmosphere, washed three times with a distilled water and ethanol mixture at a ratio 8:2, respectively, until neutral pH. Finally, the sample was dried in an oven for 8 hours at 100 °C.

Results and Discussion

X-ray diffraction analysis XRD revealed prominent carbon phases (002) and (100), as reported by (Zeng *et al.* 2020), with distinct 2θ values at 26.4° and 43° for both VR-700 and VR-700/Cu. These diffraction peaks are characteristic of graphitic carbon structures, confirming the presence of a crystalline, graphitized phase in both materials. The data showed that the incorporation of copper does not disrupt the graphitic nature of VR-700, suggesting structural integrity and enhanced conductivity suitable for electrochemical applications.

The FTIR spectrum of VR-700 reveals significant structural features. A distinct peak observed at 1540 cm^{-1} proved the presence of conjugated alkene groups (C=C), indicating the development of a partially graphitized carbon structure, as observed also in XRD. Additionally, a characteristic peak around 450 cm^{-1} suggests the presence of metal–metal bonding, which is typically associated with the formation or incorporation of metallic species within the carbon matrix. These findings imply successful integration of both organic and Cu metallic components in VR-700, enhancing its potential as a functional material in electrocatalysis, stability, surface area, and other advanced applications requiring synergistic carbon–metal interactions.

The dominant peaks obtained from XPS survey in both samples are C1s 284 eV and O1s at 531eV, peaks of P2p and S2p were seen at 133eV and 164 eV, respectively. The appearance of P2p at 132eV showed VR is successfully activated with phosphoric acid. Table 1 shows the elemental composition of both catalysts. VR-Cu-700 showed Cu2p at 933ev and CuO at 943 showing Cu(II) oxidation state. VR-700/Cu has 76.6 wt % C, whereas VR-700 has 74.35 wt % C, This shows better graphitization of VR-700. VR-700/Cu shows better phosphorus doping (8.78 wt %) than VR-700 (8.55 wt %).

Table 1—XPS Elemental composition of VR-700 and VR-700/Cu

VR-700	Weight %	VR-Cu-700	Weight %
C1s	74.35	C1s	76.69
O1s	12.03	O1s	10.91
P2p	8.55	P2p	8.78
S2p	3.77	S2p1	1.68
N1s	1.3	N1s	1.44
		Cu2p3	0.49

The TGA profile ascertains the thermal stability of both VR-700 and VR-700/Cu electrocatalysts. It was observed that VR-700/Cu exhibits improved stability compared to VR-700 without Cu doping. This improvement may be due to the presence of more moisture trapped in the VR-700 structure, which is released upon heating, as shown in Figure 2. The incorporation of copper likely contributes to a more robust structure, indicating that metal doping enhances thermal resistance and structural integrity of the electrocatalyst under elevated temperatures.

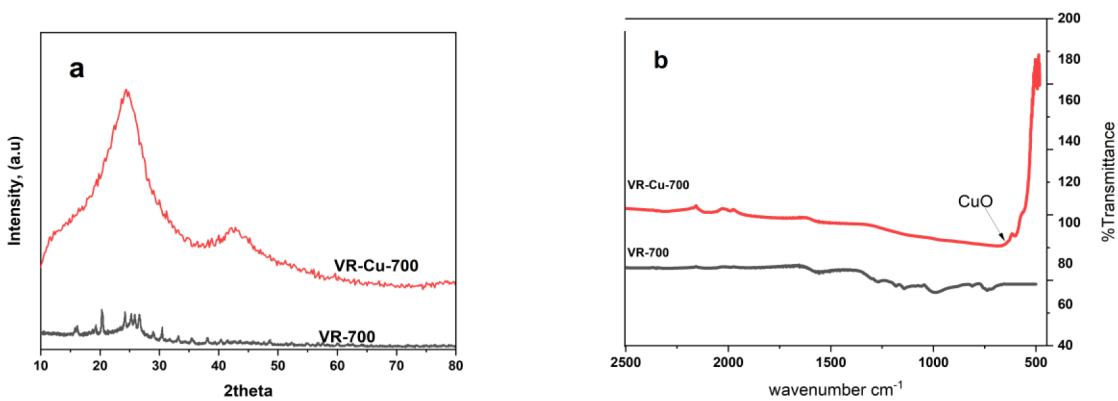

Figure 1—a) XRD pattern of VR-700 and VR-Cu-700 b) FTIR spectra of VR-700 and VR-Cu-700

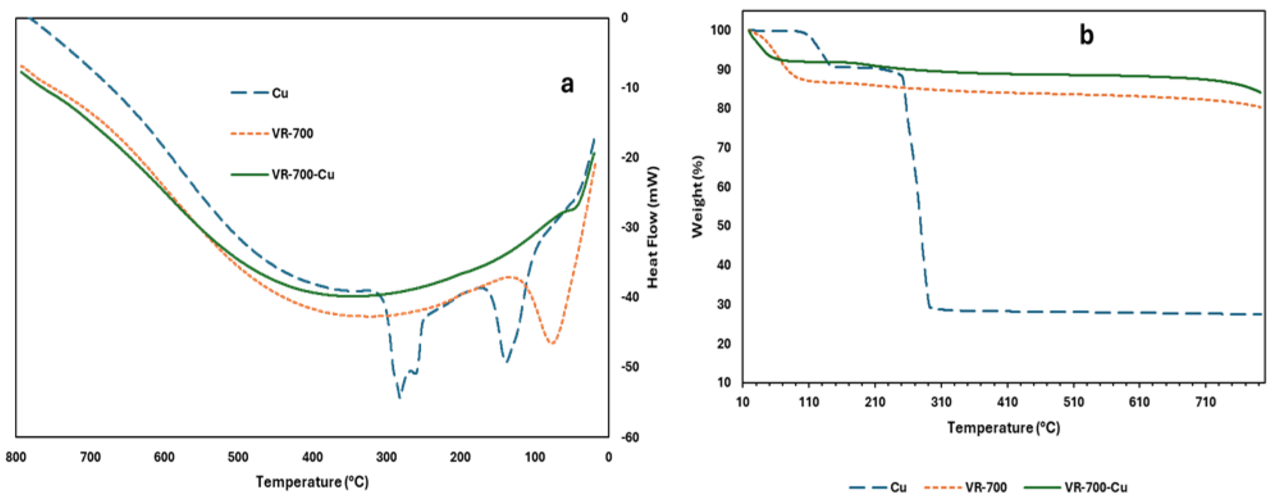

Figure 2—a) TGA profile of Cu, VR-700 and VR-Cu-700 b) DSC curve of Cu, VR-700 and VR-Cu-700

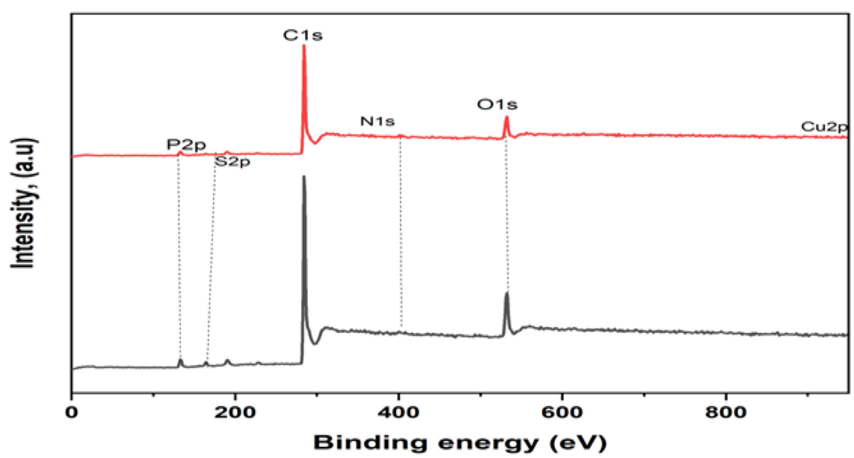

Figure 3—XPS Survey spectra of VR-700 and VR-Cu-700

Electrochemical characterization

The electrochemical performance is monitored using 3-electrode system in an alkaline medium (0.1M KOH) with rotation of 1600rpm, both VR-700 and VR-700/Cu were tested under inert atmosphere(Ar-flow) in both cases no cathodic current were observed, when VR-700 and VR-700/Cu were subjected under Oxygen flow a strong cathodic peak was observed as seen in Figure 4 and and figure 5 for VR-700/Cu and VR-700 respectively. VR-700 showed lower supercapacitance and double-layer capacitance properties compared to VR-700/Cu, as shown in Tables 3 and 4. The electrochemical surface area (ECSA) of VR-700 was found to be 92 m^2/g, while VR-700/Cu exhibited a significantly higher ECSA of 133 m^2/g. This enhanced surface area explains the superior electrochemical activity of VR-700/Cu compared to VR-700, as illustrated in the linear sweep voltammetry (LSV) results (Figure 6b), where VR-700/Cu generated a higher current under identical conditions at 1600 rpm. This confirms the enhancement of the conductivity of VR-700 after doping with Cu. Furthermore, electrochemical impedance spectroscopy (EIS) analysis confirmed faster and more efficient electron transfer in VR-700/Cu, as evidenced by smaller semicircle diameters in Figures 4d and 5d. These findings suggest improved charge transport properties due to the incorporation of copper. Chronoamperometry conducted for both electrocatalysts over 3600 seconds under continuous oxygen flow at 1600 rpm showed stable performance, further validating the structural and electrochemical stability of both materials. These results are consistent with thermogravimetric analysis (TGA), which also demonstrated their thermal stability. Overall, the data confirms that Cu incorporation enhances both the surface area and electron transfer capabilities of VR-700, making VR-700/Cu a more efficient and stable electrocatalyst for oxygen reduction reaction (ORR) and other energy conversion applications.

Table 2—Specific Capacitance of VR-700 and VR-700/Cu

Sample	Specific capacitance (F/g)					
	10mV/s	20mV/s	40mV/s	60mV/s	80mV/s	100mV/s
VR-700	3.00	2.70	2.30	2.10	2.00	1.80
VR-700/Cu	3.11	2.98	2.86	2.85	2.78	2.68

Table 3—Capacitance of VR-700 and VR-700/Cu

Sample	Capacitance (F)					
	10mV/s	20mV/s	40mV/s	60mV/s	80mV/s	100mV/s
VR-700	0.00604	0.00547	0.00477	0.00430	0.00387	0.00371
VR-700/Cu	0.00623	0.00598	0.00574	0.00571	0.00556	0.00538

Table 4—shows double layer capacitance (Cdl) and active electrochemical Surface area (ECSA) of VR-700 and VR-700/Cu.

Sample ID	Sp. Cap (F/g)	Cdl (F/cm²)	ECSA (m²/g)
VR-700	3.021	0.02799	92.6
VR-700/Cu	3.11	0.04156	133.6

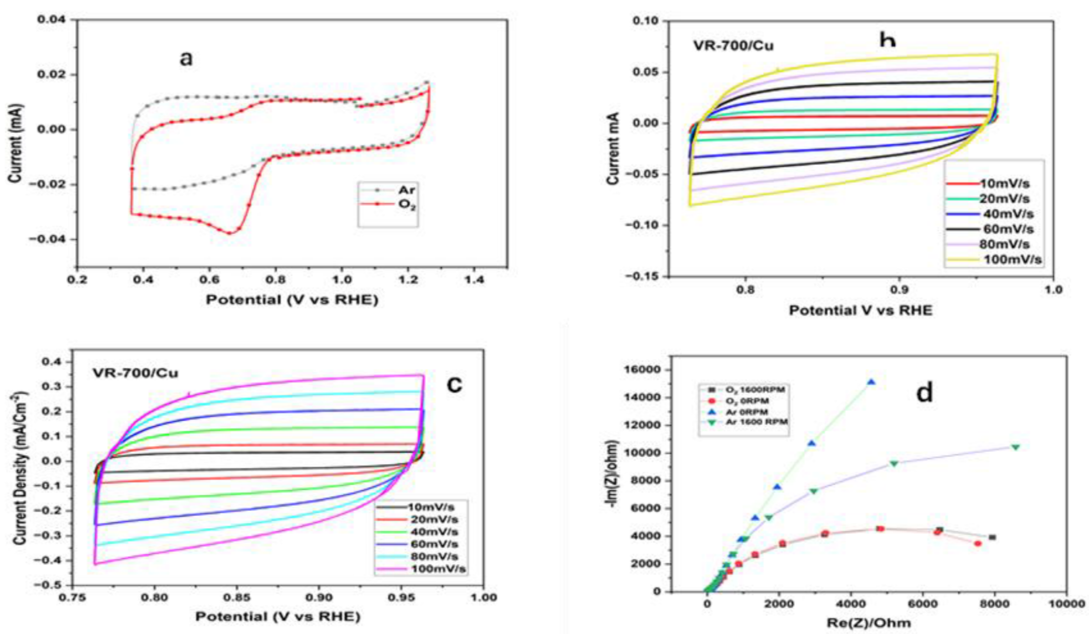

Figure 4—a) cyclic voltammetry under Ar and O_2 flow of VR-700/Cu. b) and c) cyclic voltammetry at various scan rates of VR-700/Cu d) Electrochemical impedance spectroscopy of VR-700/Cu.

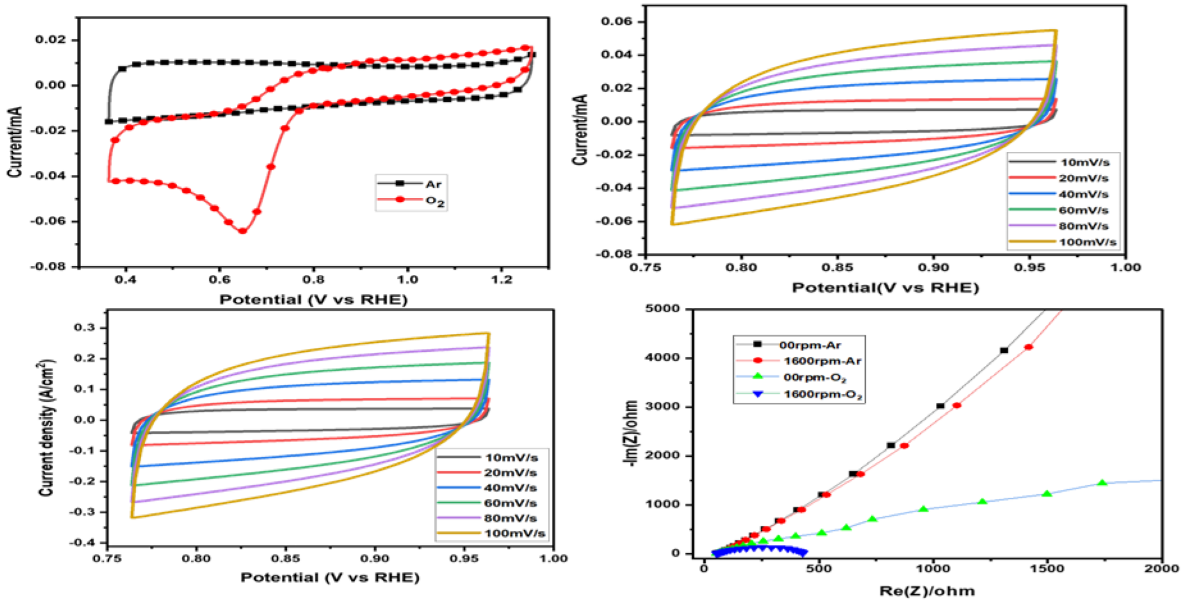

Figure 5—a) cyclic voltammetry under Ar and O_2 flow of VR-700. b) and c) cyclic voltammetry at various scan rates

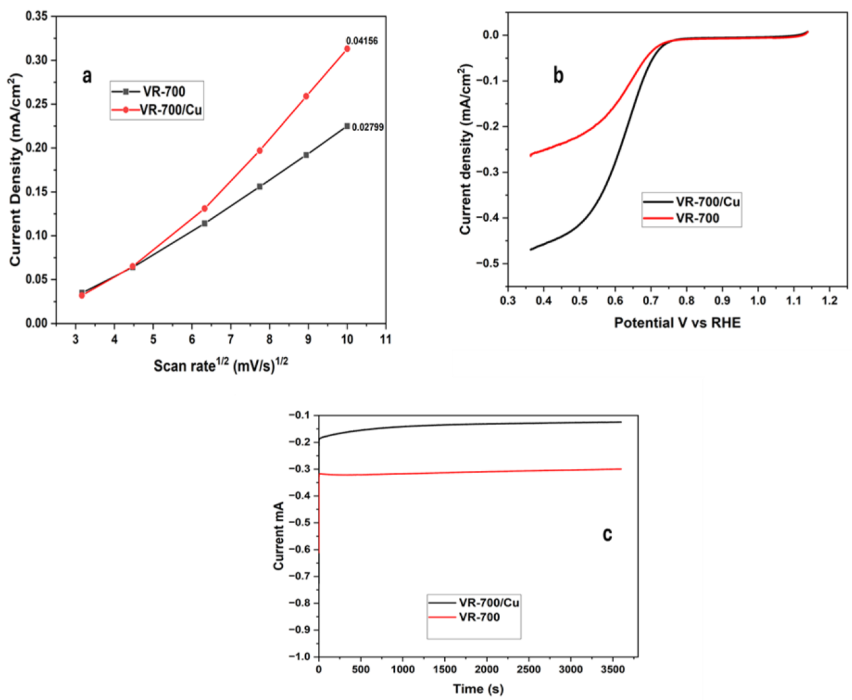

Figure 6—(a) Double layer capacitance plot of VR-700 and VR-700/Cu (b) linear sweep voltammetry of VR-700 and VR-700/Cu. (c) Chronoamperometry of VR-700/Cu and VR-700.

Deep Learning Prediction

To address the kinetic limitations of metal-air batteries, particularly at the cathode, a new electrocatalyst was designed using large language models (LLMs) to accelerate the discovery and understanding of catalytic behavior. ChemGPT (Frey *et al.* 2023) and ChemBERT (Ahmad *et al.* 2022), two state-of-the-art LLMs from hugging face adapted for chemical prediction tasks, were employed to generate novel electrocatalyst candidates and to model the underlying reaction mechanisms, respectively. In this work, we use the 151 data set as shown in Table 5. The data set was split into two 70% training and 30% testing. The training datasets were derived from SMILES representations of functional groups, doping techniques, and electrolyte media, which were processed using RDKit to ensure consistent molecular encoding. ChemGPT demonstrated strong predictive performance, generating viable electrocatalyst structures with an accuracy of 0.83. In parallel, ChemBERT, optimized to a classification accuracy of 0.80, was used to analyze and predict plausible reaction pathways. The confusion matrix in Figures 8a and b, shows similar recognition of both Minority and majority classes by ChemBERT and ChemGPT. ChemGPT generated new material by suggesting the use of material, suggesting the doping of Rhodium and Iodine in the Carbon Matrix after learning from the dataset. This integrated methodology, which combines structure-activity relationships with mechanistic predictions, offers a powerful approach for accelerating catalyst identification and improving the electrochemical performance of next-generation metal-air battery systems.

Table 5—Sample dataset for ORR mechanism prediction

S/N	Sample	Functional group	Doping	Electrolyte	Mechanism
1	C1=CC2=C(C=C1)C3=NC4=C(C=CC4=N3) C5=C2C=CN5	C1(C(C(C(C1(F)F)(F)F)F)F)	[Co]	ACIDIC	OO
2	C1=CC2=C(C=C1)C3=NC4=C(C=CC4=N3) C5=C2C=CN5	C1=CC=C(C=C1)C#N	[Co]	ACIDIC	OO
3	C1=CC2=C(C=C1)C3=NC4=C(C=CC4=N3) C5=C2C=CN5	[Cl]	[Fe]	ACIDIC	O OO
4	CN1C=CC2=C1C(=O)N(C(=O)N2)C	[O]	[Fe]	NEUTRAL	O
.					
151	CN1C=CC2=C1C(=O)N(C(=O)N2)C	[N]	[Co]	NEUTRAL	O OO

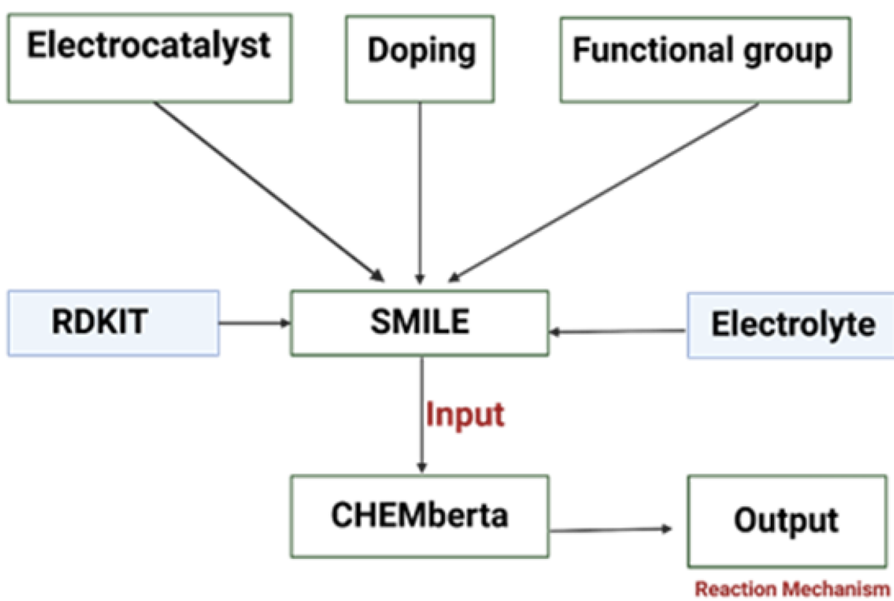

Figure 7—Scheme of deep learning model for Electrocatalyst ORR mechanism prediction

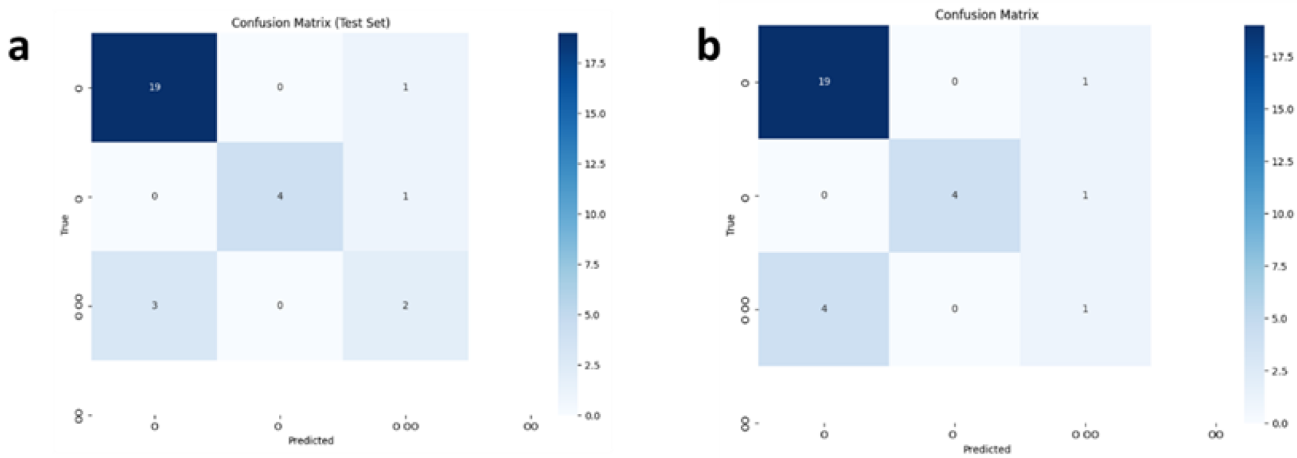

Figure 8—Confusion Matrix of a) ChemGPT b) ChembERT

Conclusion

VR-700 and VR-700/Cu were synthesized in one step using petroleum waste precursor and phosphoric acid; the latter was doped with Cu to study the effect of metallic doping and efficiency and stability. These materials have a good Electrochemical surface area and high Stability as shown in Thermo gravimetric analysis and Chronoamperometry. XPS confirmed the double-doping of both Phosphate and Cu into the structure of the electrocatalyst. VR-700/Cu showed higher activity due to a higher Electrochemical surface area. This finding, coupled with deep learning prediction of Reaction mechanism using two models, ChemBERT and ChemGPT plays a key role in the generation of new electrocatalysts for targeted applications.

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